

Week 01: What generative AI does

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Separate a model from the application that checks and delivers its output.

Context and teaching plan

A student asks whether a fictional college reimburses a late bus ticket.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Generation and prediction

A predictor assigns a label or value. A generator creates a new sequence or image. Both can make errors.

Teacher question: Where does generation and prediction appear in the weekly case, and what could go wrong if we omit it?

Model and system

The model proposes words. Retrieval, permissions, validation and people decide what the application may present.

Teacher question: Where does model and system appear in the weekly case, and what could go wrong if we omit it?

Prompt and context

A prompt states the task; context contains the supplied evidence. The model cannot reliably invent a missing policy.

Teacher question: Where does prompt and context appear in the weekly case, and what could go wrong if we omit it?

Baseline and variability

A manual answer sheet gives us a reference. Repeated responses can vary, so record both correctness and changes.

Teacher question: Where does baseline and variability appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: Policy P1 says student travel is not reimbursed.
- Change or question: Q: Is a late bus ticket reimbursed?
- Supported observation: A: No; travel is not reimbursed [P1].

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The late-bus refund: A confident assistant says a late bus ticket is reimbursed. Policy P1 says student travel is not reimbursed. Would you trust the answer?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Generation and prediction: analogy, mechanism and check

Familiar picture: A weather forecast predicts a category; a novelist writes a fresh sentence.

Familiar part	Technical meaning
Forecast label	predicted class
New sentence	generated sequence
Weather record	training examples

Where this analogy breaks: A generative model also predicts the next token internally; the distinction concerns the output task.

Actual idea: A predictor assigns a label or value. A generator creates a new sequence or image. Both can make errors.

Work it aloud: Given “Is travel reimbursed?”, a classifier could output NO; a generator could write “No, under P1.”

Ask before revealing: Can a newly worded answer still be false?

Supported response: Yes. Novel wording does not establish factual support.

Look for this misconception: Every generated sentence is creative evidence.

Model and system: analogy, mechanism and check

Familiar picture: A cook proposes a dish; the restaurant checks ingredients and allergies before serving.

Familiar part	Technical meaning
Cook	model
Ingredient sheet	policy P1
Server check	application validation

Where this analogy breaks: A restaurant check is a deliberate rule; a model does not automatically enforce permissions.

Actual idea: The model proposes words. Retrieval, permissions, validation and people decide what the application may present.

Work it aloud: The model proposes “Yes”; the application compares the claim with P1 and flags it.

Ask before revealing: Which component can stop an unsupported answer reaching the student?

Supported response: The application check using P1 can flag or block it.

Look for this misconception: The model itself guarantees the college rule.

Prompt and context: analogy, mechanism and check

Familiar picture: A librarian needs both the question and the correct shelf card.

Familiar part	Technical meaning
Request	task prompt
Shelf card	supplied context P1
Draft reply	model output

Where this analogy breaks: The shelf card helps only if it is relevant, current and actually used.

Actual idea: A prompt states the task; context contains the supplied evidence. The model cannot reliably invent a missing policy.

Work it aloud: “Answer using P1 and cite it” includes a task and source; “help me” does not.

Ask before revealing: What should the prompt say when P1 lacks the answer?

Supported response: State that the assistant should say evidence is missing.

Look for this misconception: Longer prompts always create true answers.

Baseline and variability: analogy, mechanism and check

Familiar picture: A teacher marks ten scripts against one answer key, then repeats two questions.

Familiar part	Technical meaning
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Answer key	manual baseline
Script	model reply
Repeat	variability check

Where this analogy breaks: One short sample cannot estimate performance in all future situations.

Actual idea: A manual answer sheet gives us a reference. Repeated responses can vary, so record both correctness and changes.

Work it aloud: Seven of ten replies are supported: $7/10 = 70\%$; log the three failures and two repeats.

Ask before revealing: Why record the denominator and repeats?

Supported response: The denominator shows sample size; repeats show whether outputs change.

Look for this misconception: A 70% score means the next answer is certainly correct.

New situation for discussion

The assistant repeats the answer twice, but one version invents an exception. Which answer belongs in the ten-case baseline?

Memory cue: A polished reply is a draft; trace its claim back to P1.

Guided laboratory

Run the ten-case baseline and change a single answer.

1. Read ten short questions and their reference answers
2. Run the supplied rule-based classroom stand-in
3. Mark each answer correct, incorrect or unsupported
4. Repeat two questions and compare what changed

Individual change: Change one answer to an unsupported claim and measure the score again.

Expected evidence: Ten-case answer table with one error analysis.

Checks and common error

Measure: Correct supported answers / 10 questions. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: A fluent answer is evidence that it is correct.

Correction: Fluency is a property of wording. Check the actual claim against the policy text.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Google, Develop a generative AI application.

<https://docs.cloud.google.com/docs/ai-ml/generative-ai/develop-generative-ai-application>

OpenAI, Evaluation best practices. <https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO1, LO4 | 2 hours concepts + 2 hours guided practice

The idea in plain language

A model predicts likely next pieces of text. The surrounding application chooses evidence, enforces access, and checks the result. A manual answer sheet gives a baseline. A replayed response is useful for an outage but cannot measure fresh model variability.

A worked example

Of ten policy questions, seven answers are supported, two are wrong and one should have abstained. Supported task accuracy is $7/10 = 70\%$. Repeating two recorded prompts gives two different wordings, but both may be correct. Compare meaning and evidence rather than exact spelling.

Practice before reading the answers

1. Explain generation versus prediction with one example of each. (4 suggested practice marks)

2. Seven of ten answers are correct and supported. Calculate accuracy and name a missing check. (4 suggested practice marks)

3. Why can a fluent answer be wrong? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. Predict whether an email is spam: a label. Generate a reply: new text. Both use learned patterns and both can be wrong.

2. $7/10 = 70\%$. Also check permissions, unsupported claims or repeated-run variation.

3. Likely wording is not proof of a claim. Compare the statement with the source or reference answer.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

G1 Google: Introduction to Generative AI

https://www.cloudskillsboost.google/course_templates/536

Study focus: Generation versus traditional machine learning.

G6 Google Cloud Tech: Introduction to Generative AI

<https://www.youtube.com/watch?v=G2fqAlgmoPo>

Study focus: Foundation overview. Metadata verified; full transcript not retrieved.

O1 OpenAI: Evaluation best practices

<https://developers.openai.com/api/docs/guides/evaluation-best-practices>

Study focus: Task-specific testing, repeated outputs and separate evaluation cases.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.